

SJ ASSIGNMENT 2026

Background: There is an example data in below

Building	Location Name	Location Number	Department	Capacity	Open Time
A	Floor 1	A-01			
A	Lobby Level1	A-01-Lobby			
A	Meeting Room 1	A-01-01	EFM	10	Mon to Fri (9AM to 6PM)
A	Meeting Room 2	A-01-02	FSS	50	Mon to Fri (9AM to 6PM)
A	Corridor Floor 1	A-01-Corridor			
A	Meeting Room 2	A-01-03	AVS	5	Mon to Sat (9AM to 6PM)
B	Floor 5	B-05			
B	Utility Room	B-05-11	ASS	30	Always open
B	Sanitary Room	B-05-12	EFM	10	Mon to Fri (9AM to 6PM)
B	Meeting Toilet	B-05-13	EFM	10	Mon to Fri (9AM to 6PM)
B	Genset Room	B-05-14	ASS	100	Mon to Sun (9AM to 6PM)
B	Pantry Floor 5	B-05-15			
B	Corridor Floor 5	B-05-Corridor			
...	...	...			

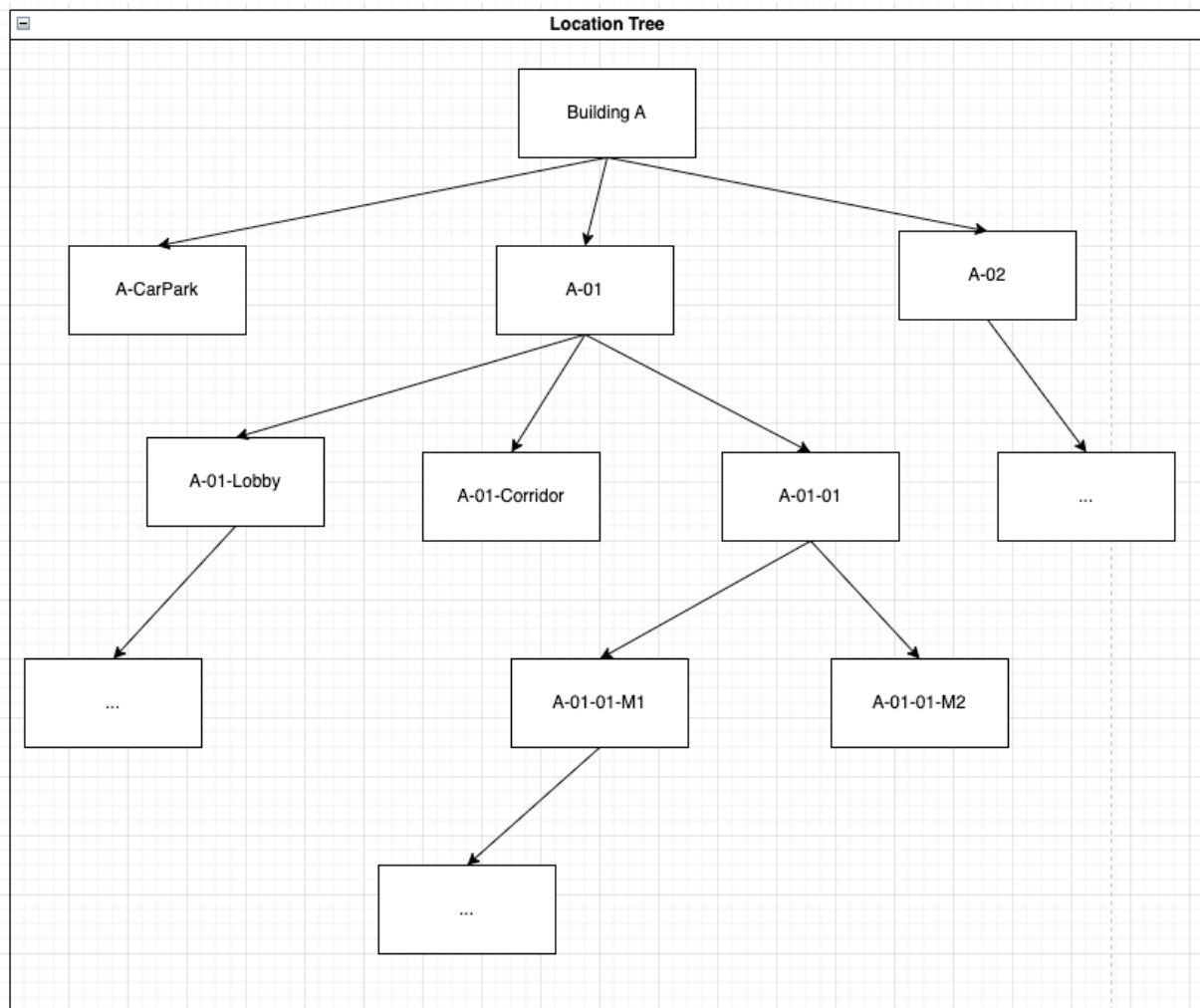
1. Project Overview

The goal of this project is to develop a RESTful API backend to manage a hierarchical tree of building locations and a room booking system. The system will handle complex location relationships (Building > Floor > Room) and enforce booking rules based on department, capacity, and operational hours.

- Primary Objective: Build a simple RESTful API for Location Management and Booking Management.
- Target Deliverable: Source code uploaded to a personal GitHub account.
- Stakeholder Contact: luc.le@coe.surbana.tech
- Duration: within 1 week.

2. Technical Stack

The project must be implemented using the following technologies



- Build with NodeJS (NestJS (<https://docs.nestjs.com>) + TypeScript + TypeORM + Postgres).
- Exception handling.
- Implement logging techniques.
- Clean code & documentation (incl. system design and database design) for your project.

3. Functional Requirements

3.1 Location Management

Users must be able to perform Create, Read, Update, and Delete operations on locations.

- Create: Add a new location node to the tree (e.g., adding a new meeting room under a specific floor).
- Read: Retrieve the location tree or specific location details.
- Update: Modify attributes like capacity or open times.
- Delete: Remove a location node.

3.2 Booking Management

The system must validate the booking requests based on the following criteria:

- Department Matching: Ensure the booking user/entity aligns with the room's assigned department.
- Capacity Check: The number of attendees must not exceed the room's defined capacity.
- Time Validation: Bookings must fall within the "Open Time" range (e.g., if a room is "Mon to Fri", weekend bookings should be rejected).

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